Case 3: Mixed Boundary Conditions (Dirichlet–Neumann) Without Source Term

1. Problem Statement

The objective of this case study is to solve the one-dimensional transient diffusion equation without a source term using the Finite Volume Method (FVM) implemented in FiPy. The problem considers a one-dimensional domain subjected to mixed boundary conditions, consisting of a Dirichlet condition at one boundary and a Neumann condition at the other.

The computational domain extends from $x = 0$ to $x = 1$. A fixed value is prescribed at the left boundary, while a zero-gradient condition is imposed at the right boundary. The transient solution is evolved in time until steady state is reached.

Unlike Cases 1 and 2, the combination of Dirichlet and Neumann boundary conditions leads to a constant steady-state solution throughout the domain. The numerical solution obtained using FiPy is validated against the analytical steady-state solution. Additional studies including transient evolution, numerical–analytical comparison, mesh convergence, timestep convergence, and error analysis are performed to assess the accuracy and stability of the numerical implementation.

2. Governing Equation

The governing equation for one-dimensional transient diffusion without a source term is

$$\partial y/\partial t = k\, \partial^2 y/\partial x^2$$

where:

• y is the dependent variable

• k is the diffusion coefficient

• x is the spatial coordinate

• t is time

For the present study:

• Diffusion coefficient, $k = 1$

• Domain length, $L = 1$

The boundary conditions are:

$y(0,t) = 1$

$dy/dx\ (1,t) = 0$

The first condition specifies a fixed value at the left boundary, while the second condition imposes a zero-flux (zero-gradient) boundary at the right boundary.

At steady state, the time derivative becomes zero:

$\partial y/\partial t = 0$

Therefore,

$d^2y/dx^2 = 0$

Integrating once:

$dy/dx = C_1$

Integrating again:

$y = C_1x + C_2$

Applying the Neumann boundary condition:

$dy/dx\ (1) = 0$ gives

$C_1 = 0$

Therefore,

$y = C_2$

Applying the Dirichlet boundary condition:

$y(0) = 1$ gives

$C_2 = 1$

Hence, the analytical steady-state solution is

$y(x) = 1$

This analytical solution is used as the reference solution for validating the numerical results obtained using FiPy.

3. Numerical Implementation

The governing diffusion equation was solved numerically using the FiPy finite volume package in Python. A one-dimensional uniform mesh was generated over the domain $0 \le x \le 1$.

Simulation parameters:

Parameter	Value
Domain Length (L)	1.0
Number of Cells (nx)	50
Cell Size (Δx)	0.02
Diffusion Coefficient (k)	1.0
Time Step (Δt)	1×10^{-3}
Number of Time Steps	2000

The initial condition was

$y(x,0) = \sin(\pi x)$

A Dirichlet boundary condition was imposed at the left boundary:

$y(0,t) = 1$

A Neumann boundary condition was imposed at the right boundary:

$dy/dx\,(1,t) = 0$

The transient diffusion equation was solved iteratively until convergence to steady state was achieved. The mixed boundary conditions result in a constant steady-state profile, representing a uniform distribution throughout the domain.

4. Results and Discussion

4.1 Transient Evolution of the Solution

The transient diffusion equation was solved using FiPy starting from the initial condition $y(x,0)=\sin(\pi x)$. As the simulation progressed, the solution evolved under the influence of diffusion and gradually approached the steady-state profile imposed by the mixed boundary conditions.

The fixed value at the left boundary continuously drives the solution toward equilibrium, while the zero-gradient condition at the right boundary prevents any net flux through the boundary. As a result, the solution gradually approaches a uniform distribution throughout the domain.

The transient evolution demonstrates how the initially non-uniform profile transforms into the final constant steady-state solution.

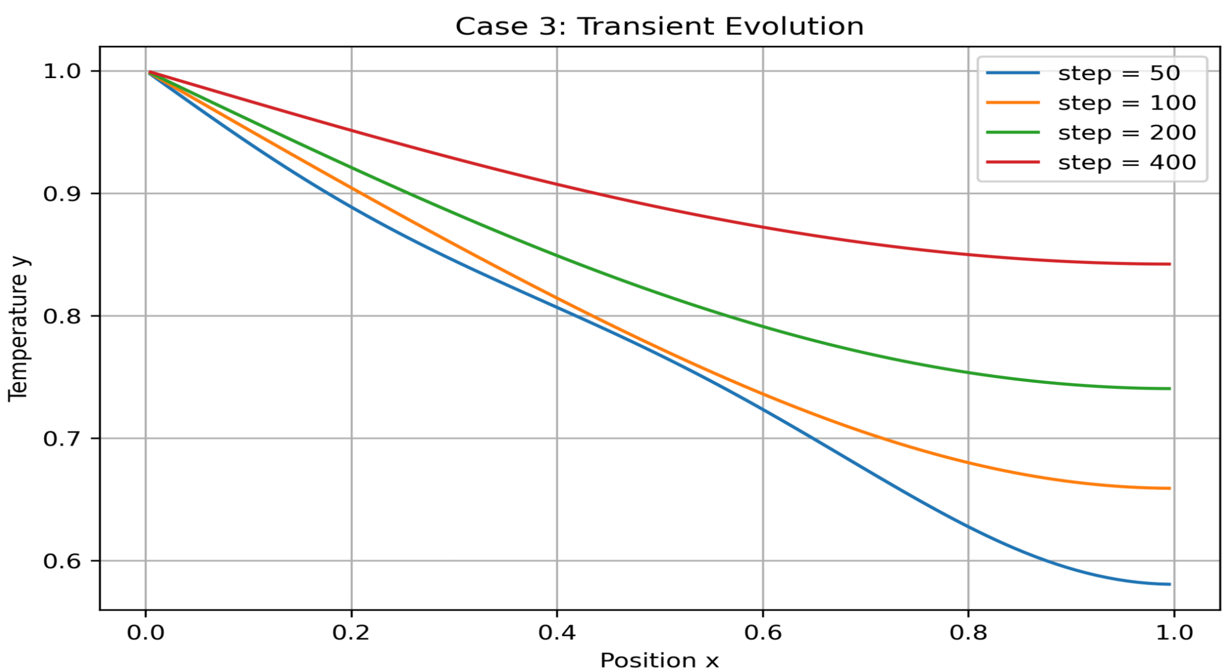

Figure 1. Transient evolution of the solution.

The solution gradually approaches a uniform steady-state profile as time progresses. Diffusion smooths the initial distribution and eliminates spatial variations throughout the domain.

4.2 Time-Step Convergence Study

A time-step convergence study was performed to evaluate the sensitivity of the transient solution to temporal discretization. Simulations were carried out using different time-step sizes while keeping all other numerical parameters unchanged.

The results indicate that as the time-step size decreases, the numerical solution converges toward a consistent solution profile. The differences between solutions obtained using smaller time steps become negligible, demonstrating temporal convergence of the numerical scheme.

The study confirms that the chosen time-step size provides a stable and accurate representation of the transient diffusion process while maintaining computational efficiency.

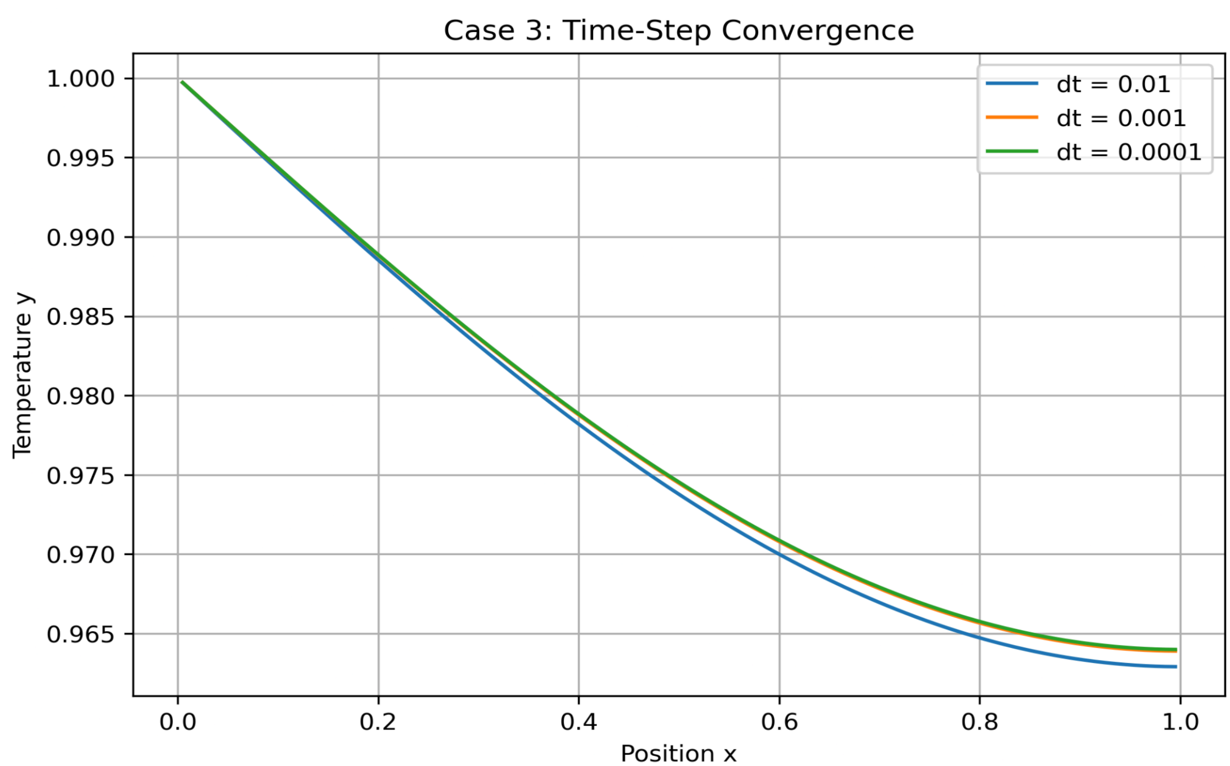

Figure 2. Time-step convergence study.

The solutions obtained using different time-step sizes show excellent agreement. This indicates that the numerical solution is largely independent of temporal discretization for sufficiently small time steps.

The overlap of the solutions obtained using $\Delta t = 0.01$, 0.001, and 0.0001 indicates temporal independence of the numerical solution.

4.3 Mesh Convergence Study

A mesh convergence study was performed to investigate the influence of spatial discretization on the numerical solution. Simulations were conducted using progressively finer meshes while maintaining the same physical parameters and boundary conditions.

As the mesh density increased, the numerical solution approached a limiting profile corresponding to the analytical solution. The differences between successive mesh refinements decreased significantly, indicating convergence of the finite volume discretization.

The results demonstrate that the numerical solution becomes increasingly independent of the mesh size as the spatial resolution is refined.

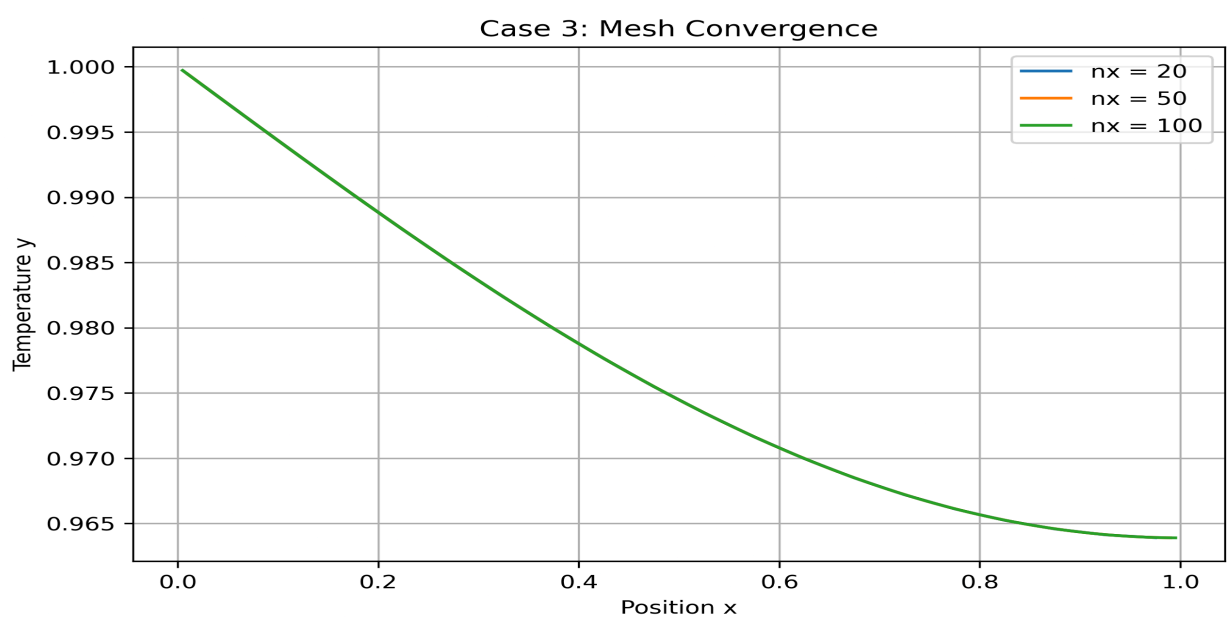

Figure 3. Mesh convergence study.

As the mesh is refined, the numerical solution converges toward a limiting profile. The decreasing difference between successive meshes confirms spatial convergence of the finite volume method.

The solutions obtained using nx = 20, 50, and 100 cells are nearly indistinguishable, confirming spatial convergence.

4.4 Steady-State Solution

After a sufficient number of time steps, the transient solution converged to a steady-state profile. Under the prescribed mixed boundary conditions, the analytical steady-state solution is given by

$$y(x) = 1$$

Unlike Cases 1 and 2, where the steady-state solutions were linear and parabolic respectively, the present case produces a constant steady-state profile throughout the domain.

The numerical solution obtained using FiPy successfully reproduced the expected uniform behavior across the entire computational domain. The resulting profile satisfies both the Dirichlet and Neumann boundary conditions.

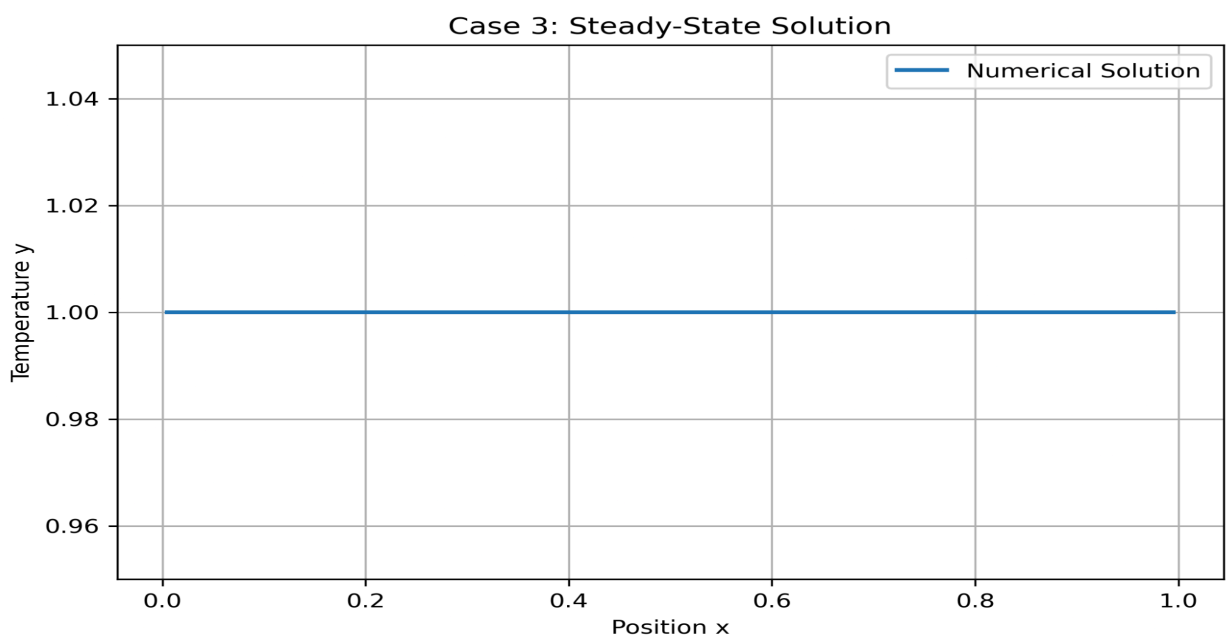

Figure 4. Steady-state solution obtained using FiPy.

The final numerical solution exhibits the expected constant profile across the domain. The solution accurately represents the steady-state behavior associated with the mixed boundary conditions.

4.5 Steady-State Verification

To verify that the solution had reached steady state, the maximum change in the solution between successive time steps was monitored throughout the simulation.

As the transient evolution progressed, the magnitude of these changes decreased steadily and eventually approached zero, confirming convergence to a stable equilibrium state.

The steady-state verification demonstrates that the selected simulation duration was sufficient to eliminate transient effects and obtain the final converged solution.

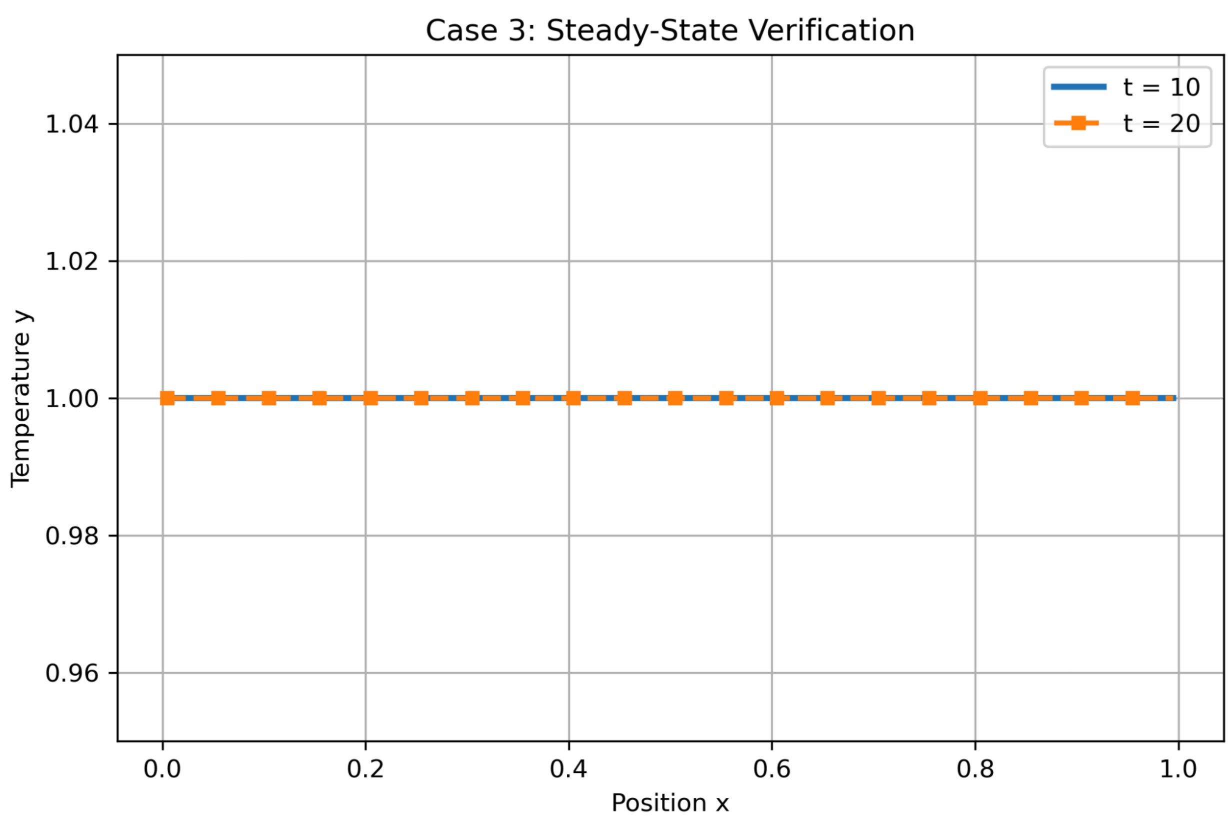

Figure 5. Verification of steady-state convergence.

The magnitude of changes between successive iterations approaches zero with time. This confirms that the numerical solution has reached a stable steady-state condition.

The profiles at t = 10 and t = 20 overlap almost completely, indicating that the solution has reached steady state.

4.6 Numerical vs Analytical Comparison

The steady-state numerical solution obtained using FiPy was compared directly with the analytical solution

$y(x) = 1$

to evaluate the accuracy of the finite volume implementation.

The comparison demonstrates excellent agreement between the numerical and analytical solutions throughout the computational domain. The numerical profile closely overlaps the analytical profile, indicating that the governing equation, boundary conditions, and numerical discretization have been implemented correctly.

The analytical solution represents a continuous function over the domain, whereas the numerical solution is computed at discrete mesh locations. The close agreement between the two validates the finite volume implementation.

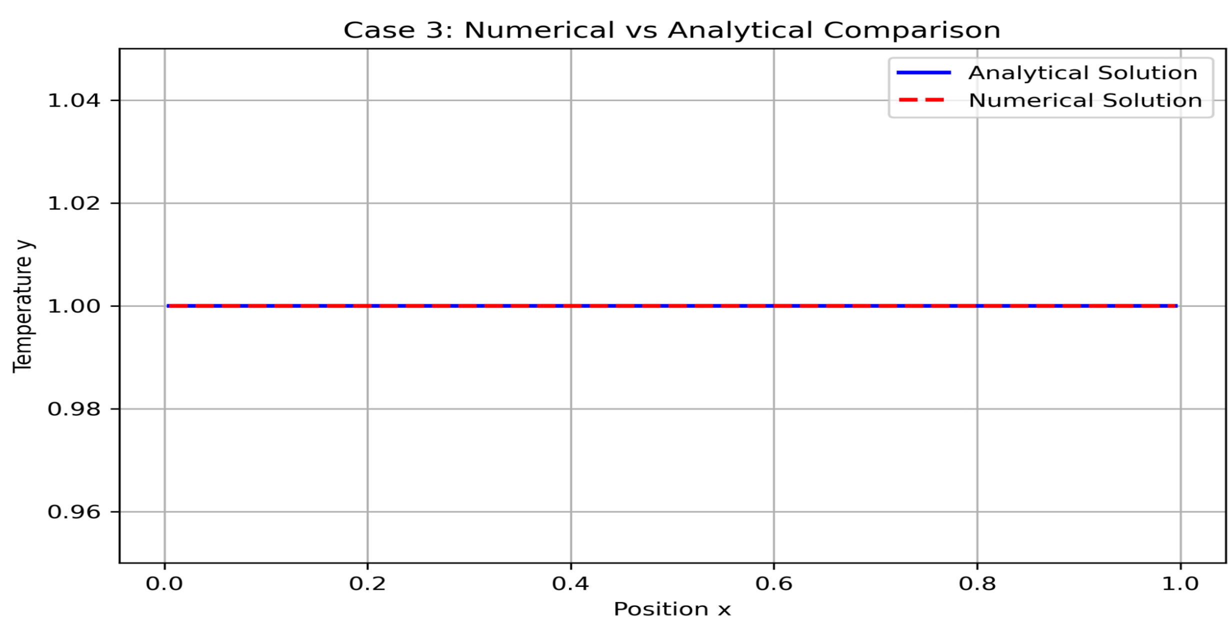

Figure 6. Comparison between numerical and analytical solutions.

The numerical and analytical solutions overlap almost perfectly throughout the computational domain. This demonstrates the accuracy of the finite volume implementation.

4.7 Error Analysis

To quantify the accuracy of the numerical solution, the difference between the numerical and analytical solutions was evaluated across the computational domain.

The low error magnitude confirms the correctness of the numerical implementation and demonstrates the capability of FiPy to accurately solve one-dimensional diffusion problems under mixed boundary conditions.

The maximum error obtained between the numerical and analytical solutions was

Maximum Error = 7.99×10^{-12}

while the L2 error norm was

L2 Error = 5.64×10^{-12}.

These error values are extremely small compared to the magnitude of the solution, indicating excellent agreement between the numerical and analytical results. The observed discrepancy arises primarily from numerical discretization and floating-point precision effects.

The maximum error is of the order of machine precision, demonstrating near-perfect agreement between the numerical and analytical solutions.

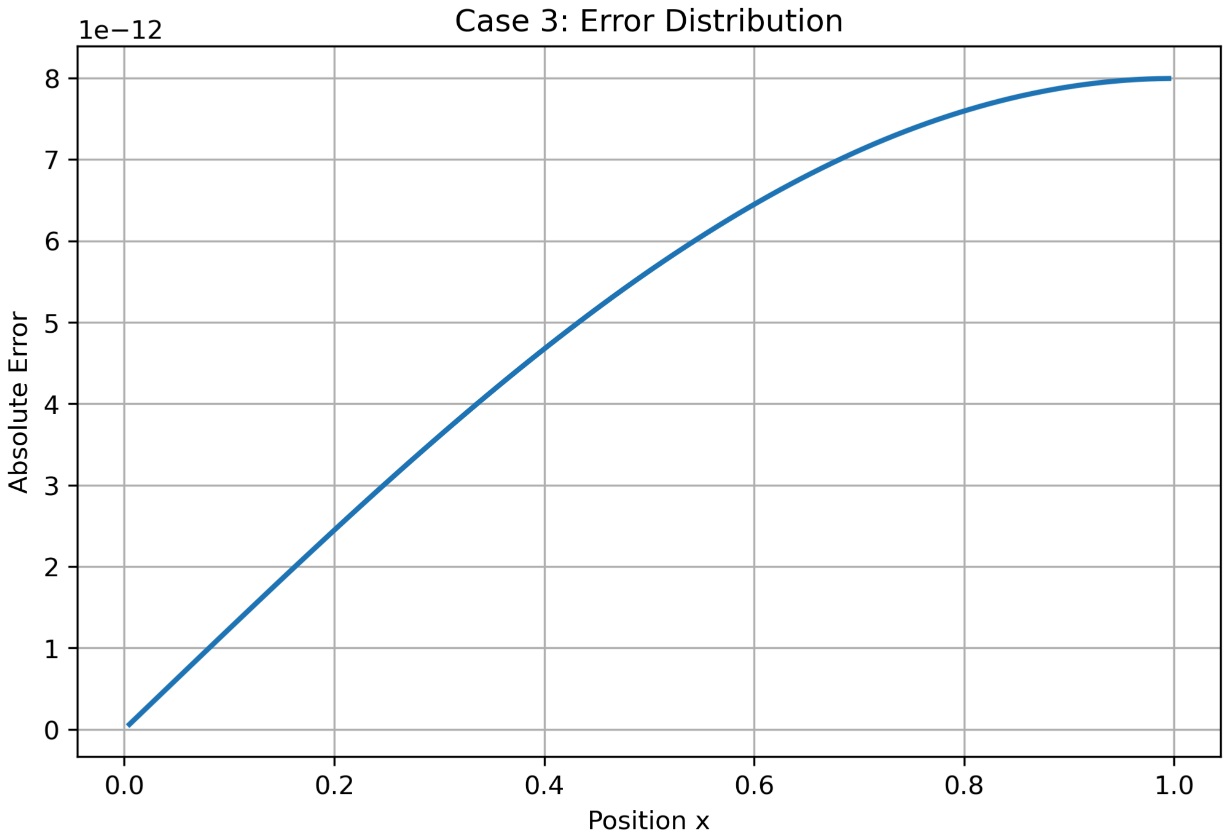

Figure 7. Error distribution between numerical and analytical solutions.

The error remains extremely small across the domain, indicating excellent agreement between the numerical and analytical solutions. The observed deviations are primarily due to discretization and numerical precision effects.

5. Conclusion

In this case study, the one-dimensional transient diffusion equation without a source term was successfully solved using the Finite Volume Method implemented in FiPy. Mixed boundary conditions consisting of a Dirichlet condition at the left boundary and a Neumann condition at the right boundary were imposed on the computational domain.

The numerical solution converged to the expected constant steady-state profile and exhibited excellent agreement with the analytical solution. Time-step convergence and mesh convergence studies confirmed the numerical stability and consistency of the solution, while error analysis demonstrated negligible deviation from the analytical result.

Overall, the study verifies the accuracy of the finite volume approach for solving one-dimensional diffusion problems under mixed boundary conditions and further validates the robustness of the FiPy implementation.